Reflection Essay: Single-View Metrology

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1 Blog Post Reflection: The Joy and Pain of Single-View Metrology

1.1 Introduction: The Magic of Homography and Cross-Ratios

Have you ever stared at a single, flat 2D photograph and wondered if you could accurately calculate the exact physical distance between two objects trapped inside the frame? Coming into this assignment for CSO-610, the idea of single-view metrology felt like a mathematical paradox. How could a camera sensor, which fundamentally discards depth information during perspective projection, retain enough geometric invariant data to let us extract real-world metrics?

The answer lies in the beautiful elegance of projective geometry—specifically, the 1D cross-ratio invariant. This assignment forced our team to bridge the gap between abstract projective geometry and applied, modern computer vision pipelines. Our goal was clear but daunting: locate a distant bicycle on the IIT Bombay campus walkway using deep learning, establish a robust 1D baseline using vanishing points, and subsequently replicate this entire experimental metrology framework using our own smartphones in a local setting in Mumbai.

1.2 The Software Stack: Building the Metrology Engine

To pull off this experiment, we engineered a clean hybrid pipeline that combined automated deep learning localization with precise manual geometric annotations. We didn't want to rely on human guesswork to anchor our target objects, so we turned to state-of-the-art computer vision tools.

Our primary engine was the `Ultralytics YOLOv8` API implemented in Python. We initially benchmarked the script using the lightweight nano model (`yolov8n.pt`), which executed lightning-fast inference times but suffered an aggressive drop-off in accuracy for small, distant objects. Because the default YOLO pre-processing architecture downscales input image tensors to a strict 640×384 footprint, the distant bicycle on the IITB walkway was squished into an unrecognizable cluster of pixels, yielding a frustrating "no detections" log.

To bypass this hurdle, we upgraded our script to load the more robust medium weights (`yolov8m.pt`) and explicitly overrode the pre-processing dimensions by forcing an uncompressed inference size of `imgsz=1280`. This optimization preserved the critical pixel density of the distant background geometry, allowing the network to comfortably isolate the bicycle (COCO class ID 1) with an extraction bounding box of `[1110.58, 2086.54, 1200.17, 2149.09]`.

For the manual layout and annotation phase, we utilized `MS Paint` to identify raw pixel coordinates. While sophisticated GIS tools exist, MS Paint allowed us to natively scrub the canvas with the mouse cursor, reading out real-time, un-aliased canvas coordinates from the bottom-left status bar. This allowed us to explicitly trace our tracking vectors and establish our reference points (A', B', C', V') without rendering artifacts.

1.3 Solving the Campus Walkway: Pixels Meet the Ground Plane

The initial phase focused on the pathway extending from the Main Building (MB) of IIT Bombay to the Arch of Knowledge. Using Google Maps, we established our true macroscopic baseline: a straight-line measurement of **140 meters** (approximated to the nearest value divisible by 10).

The mathematical heart of this assignment relies on the cross-ratio of four collinear points:

$$CR(A', B', C', V') = \frac{|A'C'| \cdot |B'V'|}{|B'C'| \cdot |A'V'|}$$

By mapping our vanishing point (V') where the parallel walkway curbs visually intersect at the horizon, we could set the real-world position of V to infinity $(Z_V = \infty)$. This brilliant geometric simplification collapses the real-world cross-ratio side to a clean relation dependent entirely on our known reference point (C)—the base of the third street pole passing parallel across the walkway.

The absolute "aha!" moment of this phase came when studying the impact of the target reference point within the YOLO bounding box. The math requires all points to lie strictly flat on a 1D line along a uniform 2D ground plane. By mapping the **bottom-center** of the bicycle box, where the tires make physical contact with the asphalt, our pixel matrices yielded a precise real-world distance of **37.3 meters** from the MB baseline.

To quantify the error, we re-calculated the cross-ratio using the geometric center of the bounding box. Because the geometric center represents a point floating physically in mid-air, it breaks the ground-plane assumption, shifting the coordinate up the canvas. This minor pixel shift translated into an immediate error of **4.8 meters**, calculating a false real-world distance of 42.1 meters. This perfectly illustrated the extreme sensitivity of perspective projection math to minor coordinate deviations.

1.4 The Local Field Experiment: Real-World Chaos and Data Gathering

The final task shifted from passive data processing to aggressive real-world field validation. We needed a custom environment featuring distinct parallel boundaries to replicate the single-view metrology pipeline. We selected a long, tiled indoor corridor section.

The data collection phase taught us a massive lesson about the environmental sensitivities of AI models. Over a 24-hour cycle, our team captured **14 unique image sets** across three distinct lighting intervals to evaluate operational boundaries:

- **Dawn (06:15 AM):** The lighting was dim and diffuse. While our human eyes could easily trace the walkway boundaries, the low-contrast environment caused YOLO's feature maps to blur, resulting in a complete failure to detect the teammate class, or logging sporadic detections with a confidence score below 0.12.
- **Morning (10:00 AM):** This was our golden hour. Crisp, directional sunlight illuminated the corridor cleanly, accentuating the high-contrast borders of the wall tiles. The YOLO medium model performed flawlessly, identifying the teammate (COCO class ID 0) with a commanding confidence score of 0.88.
- **Dusk (07:15 PM):** As twilight set in, heavy indoor shadows and ambient artificial light artifacts completely distorted the scene geometry. The model failed entirely due to sensor noise and low illumination.

To capture our final submission pair, we coordinated a three-person execution flow. I stood at the corridor baseline and captured a linear portrait of my teammate standing down the path (`firstPersonView.png`), while a third teammate stepped back to a wide angle to capture both me holding the camera phone and our subject standing down the hall simultaneously (`thirdPersonView.png`).

We utilized a physical tape measure to capture our manual ground-truth anchor: the distance from my camera position to the base of the prominent carved wooden door frame (C) was measured exactly at **3.0 meters**. By overlaying our dark blue reference line in MS Paint from the canvas edge straight through the floor-wall junction to the horizon vanishing point, we extracted our custom collinear coordinate matrix. Running these pixel distances through our cross-ratio backend mapped the teammate's physical position to **2.6 meters** away. Subtracting these vectors successfully estimated the real-world distance between my teammate and the door frame interest point to be exactly **0.4 meters**.

1.5 Conclusion: Key Takeaways from the Trenches

This project was an incredible deep-dive that balanced the rigid satisfaction of geometric proofs with the unpredictable volatility of physical engineering. We experienced the pain of model failures under dim lighting conditions and the sheer joy of watching a mathematical invariant back-calculate an indoor distance to within centimeters of a manual tape measure. We learned firsthand that deep learning models are not magic bullets; their success is deeply bound to operational variables like resolution matching, ambient illumination, and smart reference point tracking.